Lecture Week 6

Introduction to Logic (CS), Week 6

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University of Groningen

Monday-Tuesday 5-6 Oct 2026

Today's Lecture

1 Overview

2 Translating and Counting with Quantifiers

3 Proof Methods and Rules for Quantifiers

4 Conclusion

5 Additional examples

Organisational Stuff

Couple of points about the course org:

- Midterm grades will be up Tomorrow on BS
- Tutorials this week: quantifiers, translations and proofs; some selections from the midterm

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- Midterm grades will be up Tomorrow on BS
- Tutorials this week: quantifiers, translations and proofs; some selections from the midterm
- Today: tricky quantifier translations + beginning proofs
- Tomorrow lecture: bonus lecture by Gerard Renardel on Fitch proofs

So far

- Finished PL ✓
- Quantifiers: Translations ✓

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- Quantifiers: Translations ✓
- Quantifiers: Proof Rules
- Quantifiers: Formal Semantics

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- Quantifiers: Translations ✓
- Quantifiers: Proof Rules
- Quantifiers: Formal Semantics

Learning goals: At the end of this week, you are able to

- construct formal proofs for quantified sentences in FOL;

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2. Translating and Counting with Quantifiers
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5. Additional examples

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“Donkey” sentences

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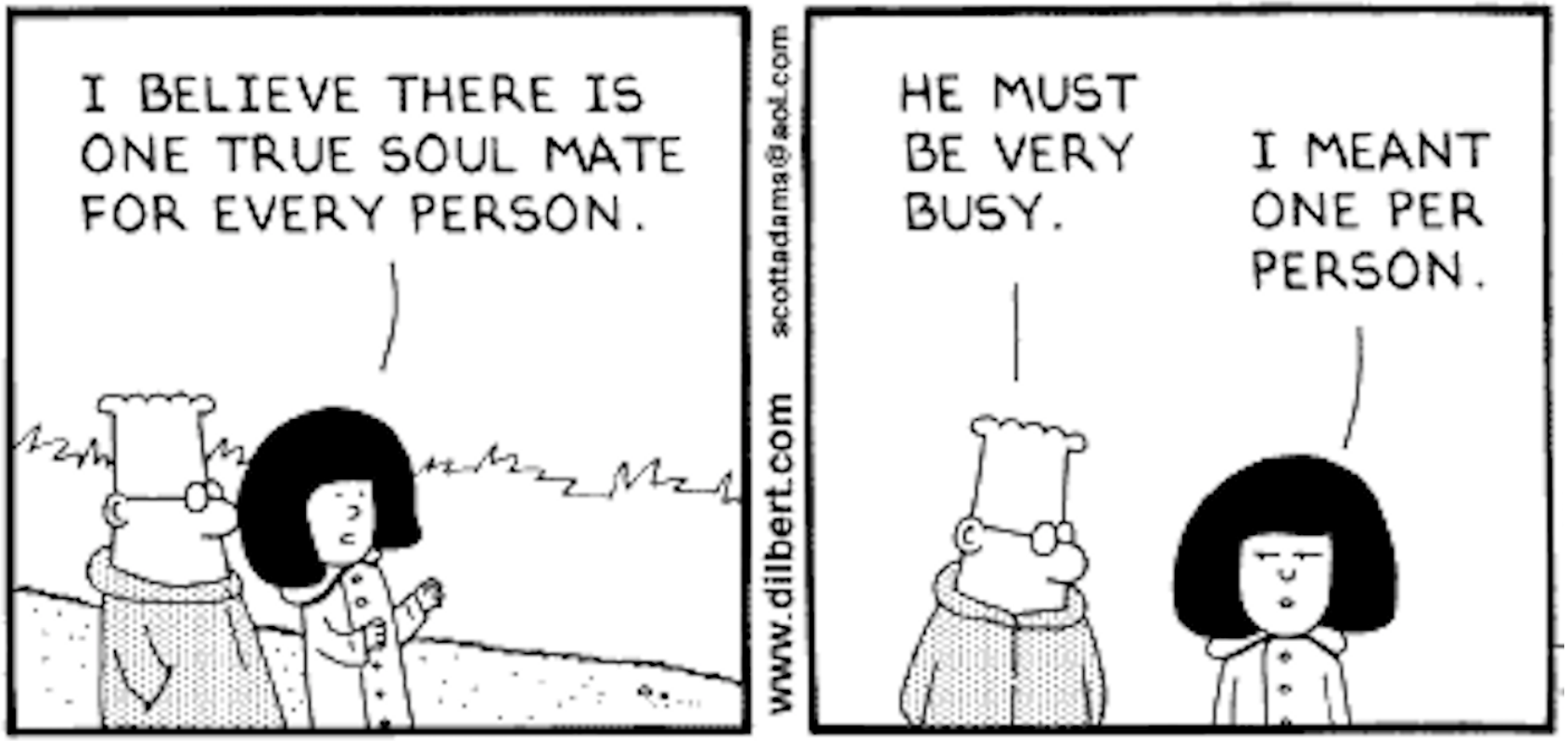
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Everyone must take good care of all dogs (s)he owns.

$\forall x \forall y((Dog(y) \wedge Owns(x, y)) \rightarrow MustTakeGoodCare(x, y))$

Mixing ∀ and ∃


I BELIEVE THERE IS ONE TRUE SOUL MATE FOR EVERY PERSON.
HE MUST BE VERY BUSY.
I MEANT ONE PER PERSON.
www.dilbert.com scottadams@aol.com

Mixing $\forall$ and $\exists$

Meaning of $\forall x \exists y A(x, y)$:

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Meaning of $\forall x \exists y A(x, y)$:

$A(x, y)$: x loves y.

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Observe: y may depend on x; different x's may require different y's to satisfy A.

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Now a *single* y must satisfy $A(x, y)$ for all x.

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Now a *single* y must satisfy $A(x, y)$ for all x.

When mixing $\forall$ and $\exists$, order matters.

Mixing $\forall$ and $\exists$

Suppose we are talking about arithmetic (DoD = integers), and consider the statements

$$A(x, y): \quad x + y = x$$

Which of the following are true?

- $\forall x \exists y A(x, y)$
- $\exists y \forall x A(x, y)$

Mixing $\forall$ and $\exists$

Suppose we are talking about arithmetic (DoD = integers), and consider the statements

$$B(x, y): \quad x + y = 0$$

Which of the following are true?

- $\forall x \exists y B(x, y)$
- $\exists y \forall x B(x, y)$

Mixing $\forall$ and $\exists$

We have:

- $\exists y \forall x A(x, y) \Rightarrow \forall x \exists y A(x, y)$, but
- $\forall x \exists y A(x, y) \not\Rightarrow \exists y \forall x A(x, y)$

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(We will make logical consequence precise in week 7.)

Example: Take $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ as the domain of discourse, then:

$$\forall x \exists y\ (x < y), \quad \text{but } \textbf{not} \quad \exists y \forall x\ (x < y)$$

Unexpressable and expressable quantifiers

Not all natural language quantifiers are expressible in FOL: “many”, “few”, etc...
(We cannot prove this formally yet)

But some others can be expressed in terms of $\forall$ and $\exists$: “exactly one...”, “at most two...”, etc

Expressing “at least 1”, “at most 1”, “exactly 1”

- *There is at least one* x *with* $A(x)$: $\exists x A(x)$

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- *There is at least one x with $A(x)$*: $\exists x A(x)$
- *There is at most one x with $A(x)$*: Reformulate: whenever two elements satisfy A, they have to be equal. In other words, *whenever $A(x)$ and $A(y)$, we have $x = y$*

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- *There is exactly one* x *with* $A(x)$*:*

  $$\exists! x A(x) := \exists x A(x) \wedge \forall x \forall y((A(x) \wedge A(y)) \rightarrow x = y)$$

  or, equivalently, $\exists x \forall y(A(y) \leftrightarrow x = y)$

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Expressing “at least 2”, “at most 2”, “exactly 2”

- *There are at least two x with $A(x)$*: ?
  Use two variables x, y and add the condition $x \neq y$:

$$\exists x \exists y (x \neq y \wedge A(x) \wedge A(y))$$

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$$\exists x \exists y (x \neq y \wedge A(x) \wedge A(y) \wedge \forall z (A(z) \rightarrow (x = z \vee y = z)))$$

or

$$\exists x \exists y (x \neq y \wedge \forall z (A(z) \leftrightarrow (x = z \vee y = z)))$$

Expressing “at least *n*”, etc

Obviously this generalizes to “at least *n*...”, and “exactly *n*...” for each concrete $n \in \mathbb{N}$.

But this *n* has to be known in advance! This suggest why we cannot use the same method for expressing “at least half of the people...” or “majority of objects...”

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Formal Proof Rules for $\forall$ and $\exists$

We expand System $\mathcal{F}$ with:

We assume:

- The domain of discourse is non-empty.
- An unbounded supply of constants/names.

Formal Proof Rules for $\forall$ and $\exists$

We expand System $\mathcal{F}$ with:

- $\forall$ Elimination
- $\forall$ Introduction
- $\exists$ Elimination
- $\exists$ Introduction

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Formal Proof Rules for $\forall$ and $\exists$

We expand System $\mathcal{F}$ with:

- $\forall$ Elimination (easy)
- $\forall$ Introduction
- $\exists$ Elimination
- $\exists$ Introduction (easy)

We assume:

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Let's start with the easy rules :-)

∀ Elimination

Also known as *universal instantiation*.

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In general: Given $\forall x P(x)$, we can instantiate x with *any closed term* c to get $P(c)$.
E.g. $P(\mathit{jacob})$, $P(\mathit{father}(\mathit{jacob}))$ etc

∀ Elimination

Formal proof rule:

⋮

i. $\forall x S(x)$ Justification (or premise)

⋮

j. $S(c)$ ∀ Elim: i

⋮

∀ Elimination

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Think of ∀ as a "big conjunction":

$$\begin{array}{l} S(a) \wedge S(b) \wedge S(c) \wedge S(d) \wedge \ldots \\ \hline S(c) \qquad\qquad \wedge \text{ Elim} \end{array}$$

Rule ∃ Introduction (Existential Introduction)

Example: Suppose:

- we have $Loves(c, c)$,
- where c is a constant (or a closed term, i.e. c refers to an object in the domain of discourse).

Then we can conclude any of:

$\exists x\, Loves(c, x)$, $\exists x\, Loves(x, c)$, $\exists x\, Loves(x, x)$, $\exists x\, Loves(c, c)$.

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In general: Suppose

- we have $S(c)$,
- where c is a constant (or a closed term, i.e. c refers to an object in the domain of discourse).

Then we can conclude $\exists x\, S(x)$ where $S(x)$ is obtained from $S(c)$ by replacing *zero or more* occurrences of c with x.

Rule ∃ Introduction (Existential Introduction)

Formal proof rule:

⋮

i. $S(c)$ Justification (or premise)

⋮

j. $\exists x S(x)$ ∃ Intro: i

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Rule ∃ Introduction (Existential Introduction)

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Think of ∃ as a "big disjunction":

$$\frac{S(c)}{S(a) \vee S(b) \vee S(c) \vee \ldots} \vee \text{Intro}$$

Example 1

Let's do an example:

$$\begin{array}{l} \forall x\, \forall y\, (p(x,y) = p(y,x)) \\ \forall x\, (p(x,a) = x) \\ \hline \exists z\, (p(z,b) = b) \end{array}$$

Think of $p(x,y)$ as $x + y$, a is zero, b is an arbitrary number.

Universal Generalization

How to prove $\forall x S(x)$?

Example: Prove: $\forall x\,(Odd(x) \rightarrow Odd(x^2))$ where the domain of discourse is the set of all integers.

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Example: Prove: $\forall x\,(Odd(x) \rightarrow Odd(x^2))$ where the domain of discourse is the set of all integers.

- Let n be an arbitrary integer.
- Assume n is odd, i.e. $n = 2k + 1$ for some integer k.

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- Let n be an arbitrary integer.
- Assume n is odd, i.e. $n = 2k + 1$ for some integer k.
- Then
  $n^2 = (2k + 1) \cdot (2k + 1) = 2k2k + 2k + 2k + 1 = 2(2k^2 + 2k) + 1$,
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- We have proved that $Odd(n) \rightarrow Odd(n^2)$
- Since n was chosen arbitrarily, we conclude that
  $\forall x\,(Odd(x) \rightarrow Odd(x^2))$.

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  $\forall x\,(Odd(x) \rightarrow Odd(x^2))$.

Universal Generalization

More generally,

- Let c be the name of an arbitrarily chosen object in the domain of discourse.
- Then . . . *[some reasoning]* . . . $S(c)$.
- Since c was chosen arbitrarily, we conclude $\forall x S(x)$.

Here, $S(c)$ is obtained by substituting all free occurrences of x with c in $S(x)$.

Universal Generalization

More generally,

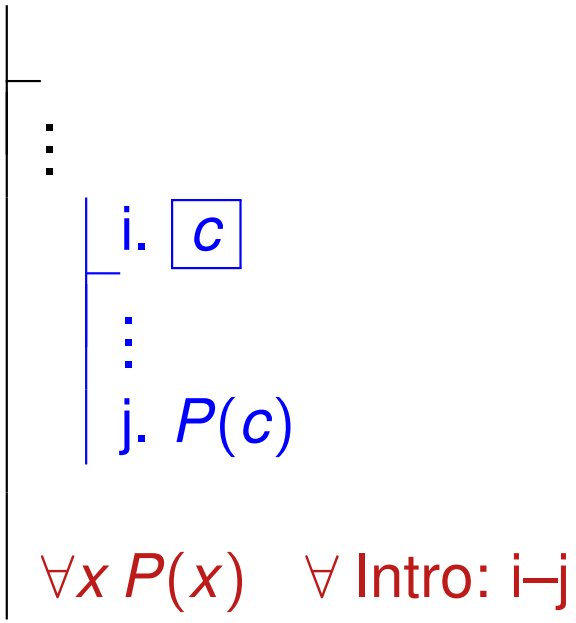
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- Since c was chosen arbitrarily, we conclude $\forall x S(x)$.

Here, $S(c)$ is obtained by substituting all free occurrences of x with c in $S(x)$.

"Object c is arbitrarily chosen" means that:
There can not be any assumptions about the properties c (only that it belongs to the domain of discourse), or any established relations between c and other objects.

Rule ∀ Introduction (Universal Generalization)

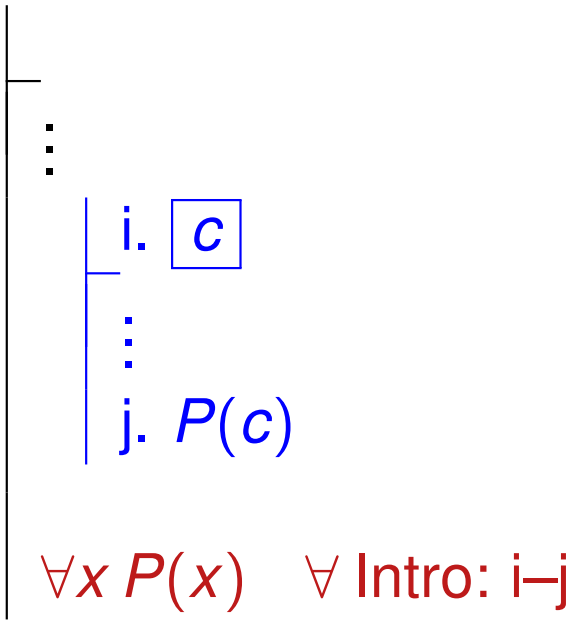
Formal proof rule:



Premise in subproof: “Let c be an arbitrary object.”.
That is, c is “fresh” and cannot occur anywhere else outside the subproof.

Rule ∀ Introduction (Universal Generalization)

Formal proof rule:



Premise in subproof: "Let c be an arbitrary object.".
That is, c is "fresh" and cannot occur anywhere else outside the subproof.

Condition: c does not occur outside the subproof where it is introduced. This warrants that c is indeed chosen arbitrarily.

Example

How to prove:

$$\forall x(x = x)?$$

Existential Elimination

Suppose we have $\exists x S(x)$.

How can we use this in a proof?
Hint: we already used it in proving $Odd(n) \rightarrow Odd(n^2)$

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$$Odd(n) \text{ means } \exists y(n = 2y + 1).$$

Since n is odd, there exists some arbitrary number m for which $n = 2m + 1$, or *Since n is odd, we can write it as* $n = 2m + 1$ *for a "fresh" m*.

Existential Elimination

In general: Suppose we have $\exists x S(x)$. How can we use this in a proof?

Let c be the name of an *arbitrarily chosen* object of the domain of discourse. Assume that $S(c)$ holds.

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Then . . . *[some reasoning]* . . . P.

So we have proven P from the assumption $S(c)$ for an arbitrarily chosen object c.

Existential Elimination

In general: Suppose we have $\exists x S(x)$. How can we use this in a proof?

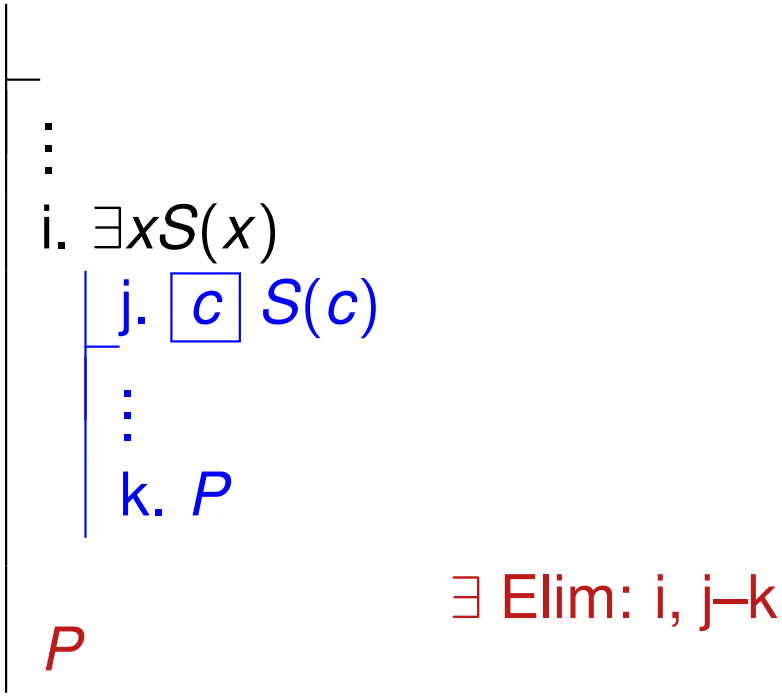
Let c be the name of an *arbitrarily chosen* object of the domain of discourse. Assume that $S(c)$ holds.

Then . . . *[some reasoning]* . . . P.

So we have proven P from the assumption $S(c)$ for an arbitrarily chosen object c.

But we know that $\exists x S(x)$, and we conclude: P.

Rule ∃ Elimination

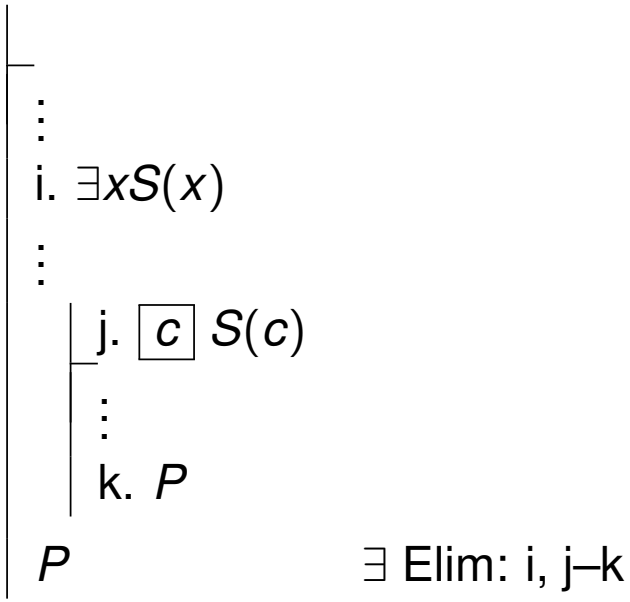


Premise in subproof: "Let c be an arbitrary object such that $S(c)$".

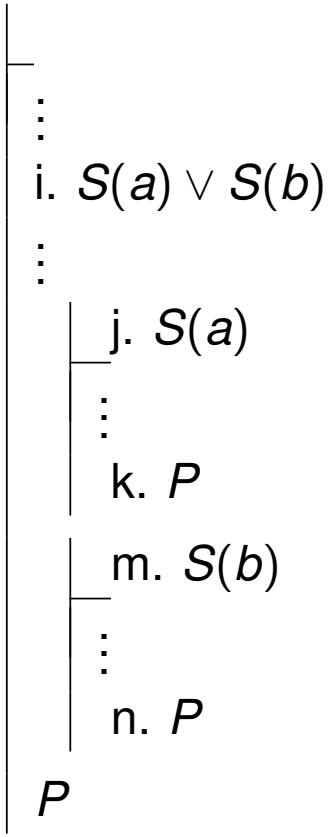
Note: You may use $S(c)$ in the subproof, not only c.

Condition: c does not occur outside the subproof where it is introduced. This warrants that c is indeed chosen arbitrarily.

∃ Elimination vs. ∨ Elimination (proof by cases)



∃ Elim: i, j–k



∨ Elim: i, j–k, l–m

Example 2

$$\begin{array}{l} \exists x\, \forall y\, P(x,y) \\ \vdots \\ \forall y\, \exists x\, P(x,y) \end{array}$$

How to obtain the conclusion $\forall y\, \exists x\, P(x,y)$?

Example 2

$$\begin{array}{|l} \exists x\, \forall y\, P(x,y) \\ \vdots \\ \forall y\, \exists x\, P(x,y) \end{array}$$

How to obtain the conclusion $\forall y\, \exists x\, P(x,y)$?

option 1 with $\forall$ Introduction as last step

Example 2

$$\begin{array}{|l} \exists x\, \forall y\, P(x,y) \\ \vdots \\ \forall y\, \exists x\, P(x,y) \end{array}$$

How to obtain the conclusion $\forall y\, \exists x\, P(x,y)$?

option 1 with $\forall$ Introduction as last step

option 2 with $\exists$ Elimination as last step

Example 2.1

$$\begin{array}{l} \forall x\ \exists y\ P(x,y) \\ \vdots \\ \exists y\ \forall x\ P(x,y) \end{array}$$

How to prove this?

Example 2.1

$$\begin{array}{|l} \forall x\, \exists y\, P(x,y) \\ \vdash \\ \vdots \\ \exists y\, \forall x\, P(x,y) \end{array}$$

How to prove this?

Trick question: it's impossible

$\forall x \exists y (x < y)$ is true for natural numbers, but $\exists y \forall x (x < y)$ is false

Example 3

Modification of our example with $+$ and 0:

$$\begin{array}{l} \forall x\, \forall y\, (p(x,y) = p(y,x)) \\ \exists z\, \forall x\, (p(x,z) = x) \\ \hline \exists z\, \forall x\, (p(z,x) = x) \end{array}$$

Today's Lecture

1 Overview

2 Translating and Counting with Quantifiers

3 Proof Methods and Rules for Quantifiers

4 Conclusion

5 Additional examples

System $\mathcal{F}$

- Reiteration rule ✓
- Introduction and Elimination rules for identity ($=$) ✓
- Introduction and Elimination rules for $\wedge$, $\neg$, $\bot$, $\vee$ ✓
- Introduction and Elimination rules for conditionals $\rightarrow$, $\leftrightarrow$ ✓
- Introduction and Elimination rules for quantifiers $\forall$, $\exists$ ✓

System of natural deduction, since close to natural reasoning in maths. There are other kinds of deductive systems (Hilbert systems, Gentzen systems, tableaux,...)

Models for ∀∃

For propositional logic we look at the semantics of *truth tables*:

- every row in a TT describes a "situation"
- the whole TT gives semantics for all of the Boolean connectives
- for ∀ ∃ TT are not sufficient: we need a way of talking about individual objects and their properties
- next week: first-order models

Today's Lecture

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Example 1a

$$\begin{array}{l} \forall x\,(\exists y R(x,y) \rightarrow P(x)) \\ \forall x\,\forall y((x = y) \rightarrow R(x,y)) \\ \hline P(a) \end{array}$$

Example 2a

Prove

$$\forall x\,(\exists y\; R(x,y) \rightarrow P(x))$$

$$\exists x \forall y\,(x = y \rightarrow R(x,y))$$

$$\vdash$$

$$\exists x P(x)$$

Example 2a

Prove

$$\forall x\,(\exists y\; R(x,y) \to P(x))$$

$$\exists x \forall y\; (x = y \to R(x,y))$$

$$\vdash$$

$$\exists x P(x)$$

How to obtain $\exists x P(x)$?

Example 2a

Prove

$\forall x\,(\exists y\; R(x,y) \to P(x))$

$\exists x \forall y\,(x = y \to R(x,y))$

$\exists x P(x)$

How to obtain $\exists x P(x)$?
Idea: existential instantiation with the fresh name we get from the second premise.